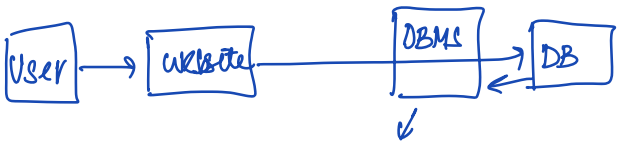
Database can store any form of structured data. → becomes information.

- Data is stored in databases electronically

Database Management System

- They are graphical user interfaces that help in retrieve, store & even manipulate it. → eg MySQL, Postgre SQL, Microsoft Access, Oracle, Mongo DB, Cassandra.
- Data is stored as binary object (0 or 1)



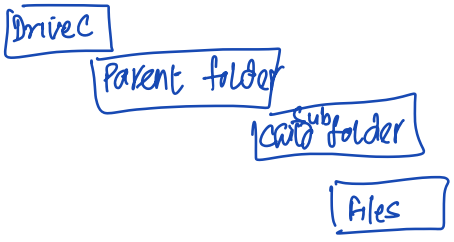
It sends data to database from user.

File Based Systems

- Alternative to DBMS – File System

  Uses file to store data on hard disk, Collections of files are stored in directories and directories may contain other files & folders.

- Each folder has its own file system embedded.



File System is a hierarchical system. They have been in existence since early 1960.

Drawback of file system-

- has redundant data (same thing - again & again) → eg we save file A.pdf - if we save again - it doesn't give error - but saves it as A1.pdf which wastes space.

eg

Name	Address
X	A
Y	B
Z	C

- There are no proper backups
- There are no query mechanisms.

  we want address of just Y → we can't get it directly from pdf w/o opening it.

- Lacks Security Access
- Data Inconsistency

  Me: Project

  My friend: Project (Edited) → But didn't share - so me - submitted outdated project.

- Metadata → (Data about data)

  $\boxed{\text{C/Parent/Sub/A.jpg}}$ → if we don't know this path, we can't access our data

A concurrency control system should provide safe access to shared information, removing the risk of collisions but in a file system, it can cause data integrity problems. Hence concurrency

is not easy to achieve in a file based system.

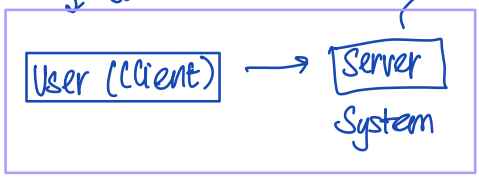
DBMS

Three types of architecture which client-server can follow →

Tier-1 Architecture

T-1
T-2
T-3

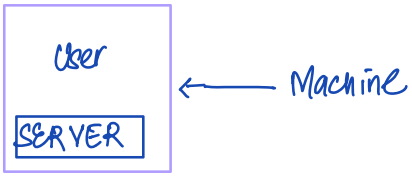
common



It could be database itself or third-party intermediary b/w user and database.

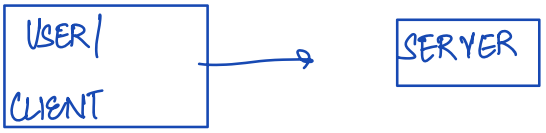
(T1)

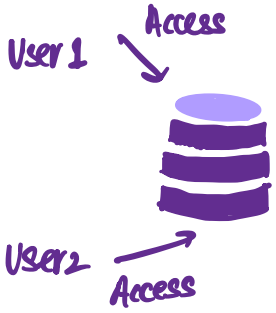
user and server reside in same machine



[Note: Way to Remember: T(1) → 1 no. of m/c]

T-2 (2 m/cs)

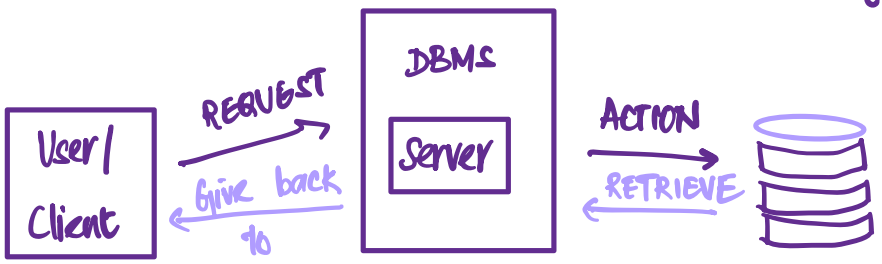




Issues → what if somebody is testing my server and they are Trojan → this may infect my server → my data goes off.

T-3 → 3 M/cs

Decoupling server & database



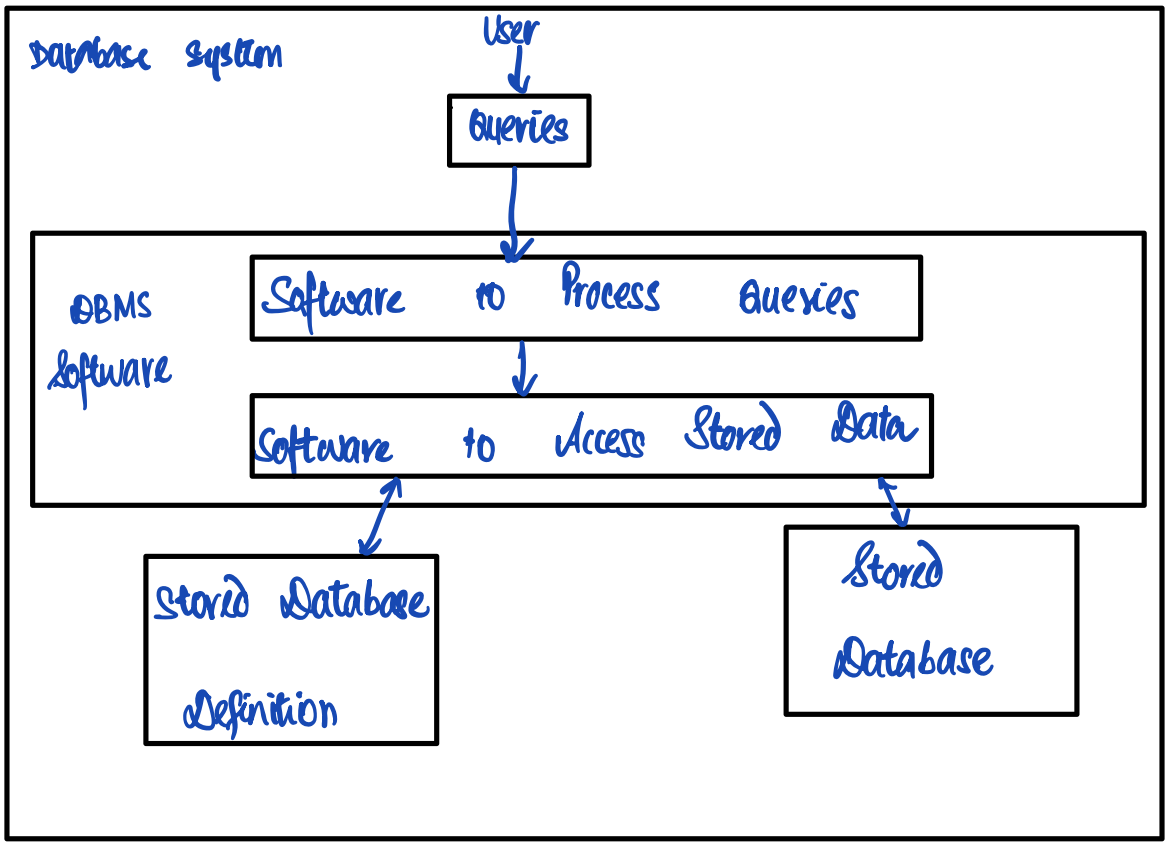
User doesn't has direct access to database.

So, if there is a virus → then server is smart enough → which stops the request then & there.

eg Ticketing System.

-ves
- → Costly
- → Training Reqd
- → Complex to use

- In DBMS, we have users who write queries using some query language, for example, SQL. (structured query language)
  - → These are processed by DBMS → eg MySQL, Oracle
  - → After queries are processed, based on results DBMS software accesses the stored data.


Database System
User
Queries
DBMS Software
Software to Process Queries
Software to Access Stored Data
Stored Database Definition
Stored Database